

Homework Assignment 2: Slab-Geometry Boltzmann Equation

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1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)v\Sigma_t t}\psi(cx, \mu, ct; 1) \quad (1)$$

(in the units $v = \Sigma_t = 1$ used by Paasschens the exponent is simply $-(1-c)t$; we keep the factor $v\Sigma_t$ so that the exponent is dimensionless).

We will do it in two parts. First, we will write down the equation that $\tilde{\psi}(x, \mu, t) \equiv \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t}\tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$. Finally we check the initial condition, which is what fixes the prefactor c .

The two equations being related are the transport equation with scattering ratio c ,

$$\frac{1}{v} \frac{\partial \psi}{\partial t} + \mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{c\Sigma_t}{2} \int_{-1}^1 \psi d\mu' \quad (2)$$

and the same equation at $c = 1$, written in its own variables (X, μ, T) ,

$$\frac{1}{v} \frac{\partial \Psi}{\partial T} + \mu \frac{\partial \Psi}{\partial X} + \Sigma_t \Psi = \frac{\Sigma_t}{2} \int_{-1}^1 \Psi d\mu' \quad (3)$$

1.1 The equation for $\tilde{\psi}$

With $\tilde{\psi}(x, \mu, t) = \Psi(cx, \mu, ct)$, i.e. $X = cx$ and $T = ct$, the chain rule gives

$$\frac{\partial \tilde{\psi}}{\partial t} = c \frac{\partial \Psi}{\partial T} \Big|_{(cx, ct)} \implies \frac{\partial \Psi}{\partial T} \Big|_{(cx, ct)} = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial t}, \quad \text{and likewise} \quad \frac{\partial \Psi}{\partial X} \Big|_{(cx, ct)} = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial x}. \quad (4)$$

Substituting these into (3) and multiplying through by c :

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} = \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (5)$$

Comparing (5) with (2), the two scattering sources already agree; the only difference is that $\tilde{\psi}$ is removed at the rate $c\Sigma_t$ instead of Σ_t . The exponential factor in (1) exists precisely to make up this deficit of $(1-c)\Sigma_t$.

1.2 Proving the identity

In this part we are going to simply substitute the RHS of (1) into the LHS of (2) and show that it satisfies the same equation. Note that $\tilde{\psi}$ is already a function of the unscaled variables (x, μ, t) , so ∂_t and ∂_x act on it directly and no further chain rule is applied:

$$\text{LHS} = \frac{1}{v} \left[-c(1-c)v\Sigma_t e^{-(1-c)v\Sigma_t t} \tilde{\psi} + c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial t} \right] + \mu c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t c e^{-(1-c)v\Sigma_t t} \tilde{\psi} \quad (6)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} - (1-c)\Sigma_t \tilde{\psi} \right] \quad (7)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} \right] \quad (8)$$

$$= c e^{-(1-c)v\Sigma_t t} \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' = \frac{c\Sigma_t}{2} \int_{-1}^1 \left[c e^{-(1-c)v\Sigma_t t} \tilde{\psi} \right] d\mu' \quad (9)$$

which is exactly (2), as required.

1.3 The prefactor c and the initial condition

Equation (2) is linear and homogeneous, so it is satisfied by any constant multiple of a solution; the PDE by itself therefore cannot determine the prefactor c . It is fixed instead by the source. For the plane-pulse Green's function the initial condition is $\psi(x, \mu, 0^+) = \frac{v}{2}\delta(x)$ for *every* value of c , and evaluating the RHS of (1) at $t \rightarrow 0^+$, using $\delta(cx) = \delta(x)/|c|$ for $c > 0$, gives

$$c e^0 \psi(cx, \mu, 0^+; 1) = c \cdot \frac{v}{2} \delta(cx) = c \cdot \frac{v}{2} \cdot \frac{\delta(x)}{c} = \frac{v}{2} \delta(x). \quad (10)$$

So the prefactor c is exactly the Jacobian of the rescaling $x \mapsto cx$ acting on the source delta, and with both the equation and the initial condition matched the identity is proved. ■

2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi \int_{|x|}^{\infty} \phi_{\text{pt}}(r) r dr \quad (11)$$

The solution taken from J. C. J. Paasschens is the point-source result $P(r, t)$ for $c = 1$. Note that Paasschens' P is normalised as a probability density, $\int P d\mathbf{r} = 1$ (his Eq. (34) summed over the collision number N), so it is a number density and the scalar flux is $\phi = vP$; we compute the density $n_{\text{pl}}(x, t)$ below and multiply by v at the end. Throughout we abbreviate

$$a \equiv v\Sigma_t t. \quad (12)$$

Applying the superposition relation to both the uncollided and the collided part of P :

$$n_{\text{pl}}(x, t) = 2\pi \int_{|x|}^{\infty} \frac{e^{-a}}{4\pi r^2} \delta(r - vt) r dr + 2\pi \int_{|x|}^{\infty} \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-a} G\left(a \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r) r dr \quad (13)$$

$$= \frac{e^{-a}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-a}}{(4\pi vt/(3\Sigma_t))^{3/2}} \int_{|x|}^{vt} \left(1 - \frac{r^2}{v^2 t^2}\right)^{1/8} G\left(a \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) r dr \quad (14)$$

$$= \frac{e^{-a}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-a}(vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \underbrace{\int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(a [1 - \xi^2]^{3/4}\right) \xi d\xi}_{\equiv I} \quad (15)$$

Note that the collided term carries $\Theta(vt - |x|)$ and not $\text{sign}(vt - |x|)$: when $|x| > vt$ the factor $\Theta(vt - r)$ annihilates the integrand for every $r \geq |x|$, so the term vanishes rather than changing sign.

2.1 Reduction of the integral I

The integral I looks forbidding, but it is elementary. Substituting first $u = 1 - \xi^2$, so that $\xi d\xi = -\frac{1}{2} du$ and the limits become $u : 0 \rightarrow u_0 \equiv 1 - x^2/(vt)^2$, and then $w = a u^{3/4}$, so that $u = (w/a)^{4/3}$, $du = \frac{4}{3} a^{-4/3} w^{1/3} dw$ and $u^{1/8} = a^{-1/6} w^{1/6}$, the fractional powers combine as $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ and

$$I = \frac{1}{2} \int_0^{u_0} u^{1/8} G\left(a u^{3/4}\right) du = \frac{2}{3a^{3/2}} \int_0^{w_0} \sqrt{w} G(w) dw, \quad w_0 \equiv a \left(1 - \frac{x^2}{(vt)^2}\right)^{3/4}. \quad (16)$$

The integrable $(vt - r)^{-1/4}$ endpoint singularity of the original integrand is removed by this change of variable, so no special quadrature is needed. Inserting I back, the prefactor collapses and the Σ_t dependence cancels completely,

$$2\pi \frac{e^{-a}(vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \cdot \frac{2}{3a^{3/2}} = \frac{e^{-a}}{2vt} \sqrt{\frac{3}{\pi}}, \quad (17)$$

so that both terms share a common prefactor and

$$\boxed{n_{\text{pl}}(x, t) = \frac{e^{-v\Sigma_t t}}{2vt} \left[1 + \sqrt{\frac{3}{\pi}} \int_0^{w_0} \sqrt{w} G(w) dw\right] \Theta(vt - |x|)}, \quad \phi_{\text{pl}} = v n_{\text{pl}}. \quad (18)$$

2.2 Closed form

With Paasschens' interpolation formula (his Eq. (36b)), $G(w) \approx e^w \sqrt{1 + b/w}$ with $b = 2.026$, the $\sqrt{w}$ produced by the substitution rationalises the square root exactly:

$$\sqrt{w} G(w) = e^w \sqrt{w + b}, \quad (19)$$

and the integral is elementary. Using $\int e^s \sqrt{s} ds = e^s \sqrt{s} - \frac{1}{2} \int e^s s^{-1/2} ds$ with $\int e^s s^{-1/2} ds = \sqrt{\pi} \text{erfi}(\sqrt{s})$ and $s = w + b$,

$$\int_0^{w_0} e^w \sqrt{w + b} dw = e^{w_0} \sqrt{w_0 + b} - \sqrt{b} - \frac{\sqrt{\pi}}{2} e^{-b} \left[\text{erfi}(\sqrt{w_0 + b}) - \text{erfi}(\sqrt{b}) \right]. \quad (20)$$

For numerical evaluation it is better to write this with the Dawson function $D(z) = e^{-z^2} \int_0^z e^{y^2} dy$, using $\operatorname{erfi}(z) = \frac{2}{\sqrt{\pi}} e^{z^2} D(z)$, which factors out the exponential growth:

$$\int_0^{w_0} e^w \sqrt{w+b} dw = e^{w_0} \left[\sqrt{w_0+b} - D\left(\sqrt{w_0+b}\right) \right] - \left[\sqrt{b} - D\left(\sqrt{b}\right) \right], \quad (21)$$

so that, since $w_0 \leq a$ and therefore $e^{-a} e^{w_0} \leq 1$, the final result contains no large or cancelling terms:

$$n_{\text{pl}}(x, t) = \frac{\Theta(vt - |x|)}{2vt} \left\{ e^{-a} + \sqrt{\frac{3}{\pi}} \left(e^{-(a-w_0)} \left[\sqrt{w_0+b} - D\left(\sqrt{w_0+b}\right) \right] - e^{-a} \left[\sqrt{b} - D\left(\sqrt{b}\right) \right] \right) \right\} \quad (22)$$

If the exact G is preferred over the interpolation, the same reduction integrates Paasschens' series term by term:

$$G(x) = 8(3x)^{-3/2} \sum_{N=1}^{\infty} \frac{\Gamma(\frac{3}{4}N + \frac{3}{2})}{\Gamma(\frac{3}{4}N)} \frac{x^N}{N!} \implies \int_0^{w_0} \sqrt{w} G(w) dw = \frac{8}{3^{3/2}} \sum_{N=1}^{\infty} \frac{\Gamma(\frac{3}{4}N + \frac{3}{2})}{\Gamma(\frac{3}{4}N)} \frac{w_0^N}{N \cdot N!}. \quad (23)$$

2.3 Checks

At the causal front $|x| \rightarrow vt$ we have $w_0 \rightarrow 0$ and the bracket tends to $\sqrt{b} - \sqrt{b} = 0$, so $n_{\text{pl}} \rightarrow e^{-a}/(2vt)$, the uncollided value alone, as it must. Integrating the result over x with the exact series G gives $\int_{-vt}^{vt} n_{\text{pl}} dx = 1$ to eight digits at $v\Sigma_t t = 0.3, 1, 3, 10, 30$, i.e. exact particle conservation for $c = 1$; with the interpolation formula the same integral gives 1.0008–1.0147, consistent with the few-percent accuracy Paasschens quotes for Eq. (36). At large t and $x = 0$ the result approaches the planar diffusion solution $(4\pi Dt)^{-1/2}$ with $D = v/(3\Sigma_t)$.

Finally, the relation proved in Q1 gives the solution for $c \neq 1$ from the $c = 1$ result above by a simple scaling and multiplication by an exponential factor,

$$n_{\text{pl}}(x, t; c) = c e^{-(1-c)v\Sigma_t t} n_{\text{pl}}(cx, ct; 1), \quad (24)$$

so the planar Green's function only ever has to be constructed once.